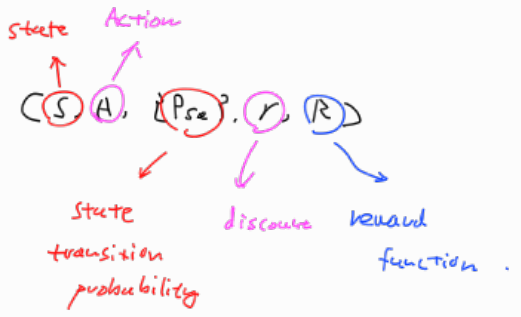
Chapter 12. Reinforcement Learning

- Supervised learning $\{(x^{(i)}, y^{(i)})\}$ — $y^{(i)}$: labels
- RL uses reward function instead.

Markov Decision Process



(i) start with state s_0

(ii) choose action a_0

(iii) MDP randomly transitions to state $s_1 \sim P_{s_0 a_0}$

$s_0 \xrightarrow{a_0} s_1 \xrightarrow{a_1} s_2 \xrightarrow{a_2} s_3 \cdots$

- total payoff $R(s_0, a_0) + \gamma R(s_1, a_1) + \gamma^2 R(s_2, a_2) \cdots$
- Question: How to maximize expected value of the total payoff

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- policy $\pi : S \to A$ (State → Action)
- value function

$$V^{\pi}(s) = E\left[R(s_0) + \gamma R(s_1) + \gamma^2 R(s_2) \cdots \mid s_0 = s, \pi\right]$$

$$= R(s) + \gamma \sum_{s' \in S} P_{s\pi(s)}(s') V^{\pi}(s') \quad \text{i.e. Bellman Equation}$$

$$= R(s) + \gamma E_{s' \sim P_{s\pi(s)}}\left[V^{\pi}(s')\right]$$

$|S|$ linear equations in $|S|$ variables

- optimal value function

$$V^*(s) = \max_{\pi} V^{\pi}(s)$$

$$= R(s) + \max_{a \in A} \gamma \sum_{s' \in S} P_{sa}(s') V^*(s')$$

Note. for every { state s, policy π }

$$V^*(s) = V^{\pi^*}(s) \geq V^{\pi}(s)$$

- optimal policy for all states

$$\pi^*(s) = \operatorname{argmax}_{a \in A} \sum_{s' \in S} P_{sa}(s') V^*(s')$$

① finite state/action spaces

② known { transition probability $\{P_{sa}\}$
reward function R

Value Iteration

Step 1. Find V^* (value func) with Bellman Equations

Note. { sync update
async update.

- init $V(s) = 0$
- $\forall s \in S$ update $V(s) := R(s) + \max_{a \in A} \gamma \sum_{s'} P_{sa}(s') V(s')$

↳ converge to V^*

Step 2. Find $\pi^*(s)$ (policy)

- $\pi^*(s) = \arg\max_{a \in A} \sum_{s' \in S} P_{sa}(s') V^*(s')$

Policy Iteration.

- init π
- { $V := V^\pi$ by linear system solver.
  $\pi(s) = \arg\max_{a \in A} \sum_{s'} P_{sa}(s') V(s')$

⇒ converge to V^*, π^*

Note.

- small MDP ⇒ Policy Iteration
- large MDP ⇒ Value Iteration.

How to estimate state transition probability
rewards

(i) $P_{sa}(s') \approx \frac{\#(s,a,s')}{\#(s,a)}$

(ii) $R(s) \approx$ average reward observed in s

$s_0^{(1)} \xrightarrow{a_0^{(1)}} s_1^{(1)} \xrightarrow{a_1^{(1)}} \cdots$

$s_0^{(2)} \xrightarrow{a_0^{(2)}} s_1^{(2)} \xrightarrow{a_1^{(2)}} \cdots$

Example : model learning + Value Iteration.

- execute π ⇒ update P_{sa} and R
  (transition probability, reward)
- value iteration ⇒ update V
  ⇒ update π

Note.

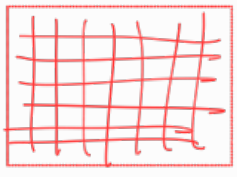
during value iteration,

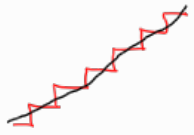
init V with previous solutions

Continuous State MDPs.

Discretization

문제 ① naive representation for V^* and π^*





Smooth 함수이나 부정확.

문제 ② curse of dimensionality

of states = k^d

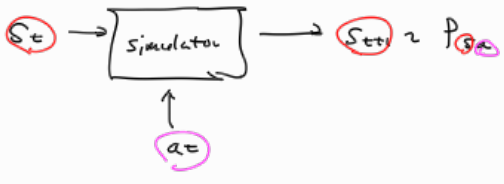
⇒ used in low-dim (1D/2D)

($\bar{S}$, A, {P_{sa}}, γ, R)

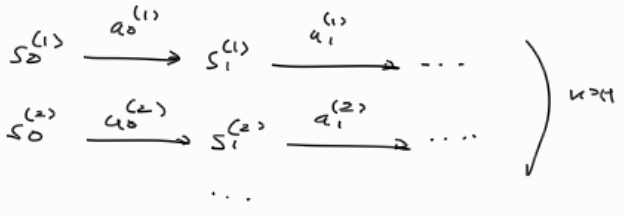
state $\bar{s}$

action $\pi^*(\bar{s})$

Value Function Approximation.



data : n state sequences



- linear model $s_{t+1} = As_t + Ba_t$

⇒ { deterministic
  { stochastic : $As_t + Ba_t + \varepsilon_t$, $\varepsilon_t \sim N(0, \Sigma)$

- $\arg\min_{A,B} \sum_i \sum_t \left\| s_{t+1}^{(i)} - (As_t^{(i)} + Ba_t^{(i)}) \right\|_2^2$

  i ↓ State sequence

Fitted Value Iteration

$$V(s) = R(s) + \gamma \max_a \mathbb{E}_{s' \sim P_{sa}} [V(s')]$$

$$= R(s) + \gamma \max_a \int_{s'} P_{sa}(s') V(s') ds'$$

Note.

- approximate over finite states $s^{(1)} \dots s^{(n)}$
- S is continuous
  A is small & discrete.

Fitted Value Iteration.

- sample n states $s^{(1)} \dots s^{(n)}$
- init $\theta = 0$
- for $i = 1, \dots, n$ (states)
  - for $a \in A$ (actions)
    - sample $s_1', \dots, s_k' \sim P_{s^{(i)}a}$
    - $q(a) = R(s^{(i)}) + \gamma \sum_j \left(\frac{1}{k}\right) V(s_j')$
  - $v^{(i)} = \max_a q(a)$
- $\theta = \arg\min_\theta \frac{1}{2} \sum_i (\theta^T \phi(s^{(i)}) - v^{(i)})^2$

Note.

- Use Supervised Learning Algorithm like linear regression

  i.e $V(s) = \theta^T \phi(s)$
- does not always converge

  ⇒ but it often does
- use $k = 1$ in deterministic MDP.

  (바둑, 체스)

- value function V
- policy $\pi(s) = \arg\max_a \mathop{E}_{s' \sim P_{sa}} [V(s')]$

(i) sample $s_1', \dots, s_k' \sim P_{sa}$

next states

(ii) if $s_{t+1} = f(s_t, a_t) + \varepsilon_t$

f: deterministic function

ε_t: zero-mean Gaussian Noise.

만약 노이즈 ε_t가 충분히 작다면? ⇒ $\pi(s) = \arg\max_a V(f(s,a))$

- just set $\varepsilon_t = 0$, $k = 1$
- $\mathop{E}_{s'}[V(s')] \approx V(\mathop{E}_{s'}[s']) = V(f(s,a))$
- $s' \sim P_{sa}$

Note. 1차 테일러 전개.

- $V(s') \approx V(\mu) + V'(\mu)(s' - \mu)$
- $E[V(s')] \approx E[V(\mu)] + E[V'(\mu)(s' - \mu)]$

$E[V'(\mu)(s' - \mu)] = 0$

Step 1. Policy Iteration.

(i) $V := V^{\pi}$ (linear system solver)

(ii) $s \in S \Longrightarrow \pi(s) := \arg\max_{a \in A} \sum_{s'} P_{sa}(s') V(s')$

Step 2. Apply Value Iteration

(i) Value Evaluation

repeat k

$\Rightarrow s \in S \Rightarrow V(s) := R(s) + \gamma \sum_{s'} P_{s\pi(s)}(s') V(s')$ $\Big]$ $=: VE(\pi, k)$

(ii) Policy Improvement

$s \in S \Longrightarrow \pi(s) := \arg\max_{a \in A} \sum_{s'} P_{sa}(s') V(s')$

